

Alternating Current Electric Generator Machine Inverters in a Parallel Power Sharing Connection

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ABSTRACT This paper covers the theoretical background as well as chosen selected experimental results obtained for two AC current generators connected to voltage source inverters working in parallel. Inverters were connected in parallel via their intermediate direct current circuits by means of auctioneering diodes which select higher voltage source, and then directly to a DC power network. The selected real-time control algorithm for a digital signal processor and a field programmable gate array was used to achieve the proper excitation of two different types of alternators: a squirrel-cage generator denoted as and a self-excited synchronous generator. Another task for the excitation system was to maintain the proper level of direct current link voltage and stable power distribution between both generators rotating at changing speeds. The windings of alternators were fed with the use of so-called machine side inverters. The asynchronous generator was a voltage source inverter driven with the use of a sensorless multiscalar algorithm based on the current machine model, whereas the self-excited synchronous generator worked on the field oriented control principle. To prove the robustness of the chosen algorithm, different types of load were applied while during the parallel work of the generators and inverters.

INDEX TERMS Space vector pulse width modulation, power distribution, smart grids, semiconductor diodes, signal processing algorithms, AC-DC power converters.

I. INTRODUCTION

Electrical grid DC systems have long been the standard in the telecommunications industry [1] and on naval ships, as the DC-ZEDS system [2]. DC systems can be used also in a small grid for home use and residential applications [3], [4] where power quality and reliability can be improved. This, of course results in improvements in electrical conversion efficiency. It is worth noting that the direct current systems are suitable for the interconnection of alternative energy sources and energy storage devices (batteries, ultracapacitors, etc.) aimed at improving the reliability of distributed generation systems. A DC based electrical grid has none of the disadvantages of alternating current systems, such as synchronization issues and active/reactive power flow which makes a DC system useful in situations where reliability and power quality are important. To improve redundancy, a direct current system should consist of multiple electrical generators. As shown

in Fig. 1, the system presented here uses two independent generators characterized by varying rotational speeds.

The use of independent electrical generators provides multiple configuration options for supplying power to the ship's grid: power can be supplied simultaneously, sequentially, or from one bus exclusively [2]. Power converters enable simultaneous operations from one or more generating sets, but they can be complicated to operate. Cycling through all available voltage sources but drawing power from only one at a time requires persistent switching and draws discontinuous current, which may excite system resonances and cause voltage oscillations.

Although such systems are widely known and described [5], [6], very little information is available concerning methods of uneven power distribution in variable speed generators connected to a grid with means of power electronics devices. This article presents the results of two different types of generators working in parallel and feeding the DC side of the network through auctioneering diodes [7].

In the classic ships power plant designs, an alternating current electrical grid is used, and the main switchboard is

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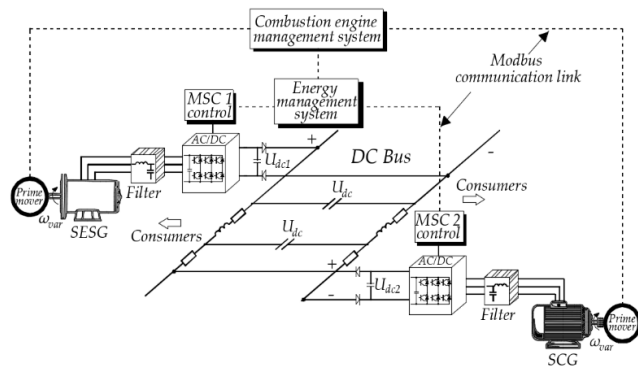


FIGURE 1. DC grid system based on a self-excited synchronous generator (SESG) and an asynchronous squirrel-cage generator (SCG) connected to the intermediate circuits of machine inverters intermediate circuits.

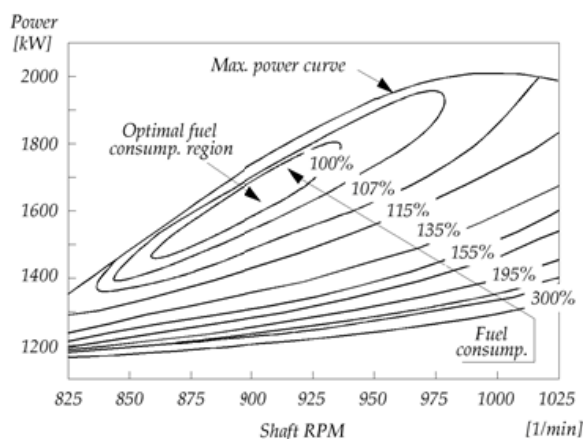


FIGURE 2. Map of combustion engine optimal fuel consumption.

the point where all electrical sources and loads are tied up together. The main reason for using the presented system is advantage of fuel savings in generator – combustion engine systems working at varying rotational speeds. Commercial onboard DC systems show the savings reaching up to roughly 27% in comparison to constant RPM working gensets [8]. The advantage of the system lies in maintaining the speed of the combustion engine within the optimal fuel consumption region range, depending on mechanical load [9].

As opposed to a “classical” electric plant, available power does not depend solely on the rotational speed of the alternators but thanks to electrical conversion in machine side converters the electrical power can be obtained by combining alternator varying angular speed and active current controlled by machine inverter. In the most of similar real-world applications, the generators used are of the self-excited synchronous type, which generate alternating current, to be later rectified subsequently by controlled SCR’s [10].

The active load sharing system presented in the paper utilizes the intermediate circuits of inverters connected via auctioneering diodes, whereas power distribution between generator sets is controlled by limiting the currents of machine-side inverters and voltages of DC

intermediate circuits. The external circuits contain resistive and capacitive loads, but transmission lines can be also considered as parallel resistances connected insulation and cables to hull capacitances. Although these quantities have influence on short-circuit and leakage current values are not discussed in presented paper.

The similar systems are installed onboard of specialized commercial vessels and indicate the benefits as follows:

- fuel savings and reduced combustion engines gas emission,
- savings of weight and space occupied by electrical devices,
- possible use of energy storage devices resulting in further efficiency increase,
- no synchronization issues,
- no active and reactive power flow and distribution problems,
- simple integration with navigational systems and main propulsion motors via digital communication.

There are also some disadvantages of using this type of power system which are arising from complicated devices arrangements, excessive heat dissipation, electromagnetic interferences and distortions but with improving technology most of these will be avoided in the future.

A. DESCRIPTION OF SYNCHRONOUS GENERATOR PROPERTIES AND THE MACHINE INVERTER CONTROL METHOD

The principle of the synchronous generators control with independent excitation winding is well known, involving the frequency and voltage control by means of the active and reactive power adjustment. The reactive power is commanded and controlled by means of the DC current excitation winding and automatic voltage regulator, while the active power is provided by the mechanical drives such as combustion engines. In most cases, the two control loops operate separately with the use of the RPM’s governor and voltage regulator. This kind of operation may be considered as a scalar control procedure, which disregards some phenomena, i.e., the coupling effect between the electrical axis in the synchronous generator [11]. One of the many suitable for this purpose methods, is based on the field orientation principle of vector control. Due to the principles of vector control, AC machine performance during transient operation modes comes quite close to that of direct current machines. The mathematical background of the dynamic model and vector control of an AC machine is illustrated by the space-phaser theory [12], [13]. The rotor flux oriented synchronous machine model is like a shunt excited direct current machine. It is suitable for the simulation of the synchronous generator operation however the control is implemented by means of a field oriented model, considering the resultant stator flux. This model suggests an analogy with a compensated DC machine, which enables the independent control of the two variables that produce the machine torque [12]–[14]. Fig. 3 depicts the stator field

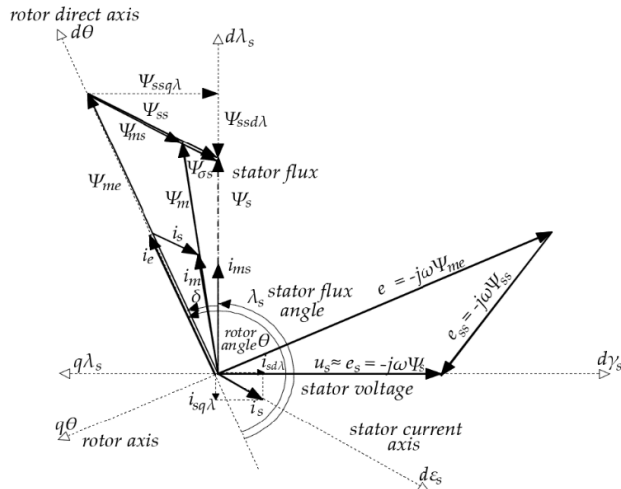


FIGURE 3. Diagram of a synchronous generator with leading stator current, and the stator field oriented space phasors of the stator current.

oriented components of the stator current. In this figure,

$$i_s = i_{sd\lambda} + j i_{sq\lambda} \quad (1)$$

whereas the armature winding flux equals:

$$\Psi_{ss} = \Psi_{ssds} + j \Psi_{ssq/s} = L_{md} i_{sd\lambda} + j L_{mq} i_{sq\lambda} \quad (2)$$

where λ_s is the angular position of the resultant stator flux Ψ_s .

In generating mode operation, the quadrature component of the flux $\Psi_{ssd\lambda}$ (see Fig. 3), which controls and determines active power and intermediate circuit direct voltage, is negative due to the reversed active energy flow. Flux denoted as $\Psi_{ssq\lambda}$, due to its demagnetizing character, has also a negative value and corresponds to the generated reactive power. The exciting current influences on an active current component, which in turn produces the torque. Because of decoupled control, the longitudinal component of the field oriented exciting current contributes solely to the magnetizing phenomena in the self-excited synchronous machine. Since the stator resistance can be omitted due to its low value, the triangle of the machine powers becomes like that of the stator currents. In such a case, active and reactive power can be controlled independently by means of active and reactive stator current components control. To have a control of the power flow, the stator current must be oriented according to the magnetizing direction of the resultant stator flux $\Psi_{ssq\lambda}$ and implemented by means of a Clarke and Park (α - β and d-q) transformations. To achieve simple and robust control for a synchronous generator, some simplifications regarding the produced electromagnetic torque must be considered. In a constant torque angle control strategy, the d axis current is kept at zero, while the vector current is aligned with the q axis. This method ensures that the torque angle equals to 90° . The torque equation for an SESG, considering both i_{sd} and i_{sq} currents, is the one as follows in:

$$T_e = \frac{3}{2} p [\Psi_m i_{sq} + (L_d - L_q) i_{sq} i_{sd}] \quad (3)$$

where: p is the number of pole pairs, Ψ_m is field winding flux, and L_d and L_q are values of the stator self inductances in the d-q plane. After substituting the d-q currents in equation (3) and after applying some simplifications, the torque equation becomes:

$$T_e = \frac{3}{2} p \Psi_m i_{sq} \quad (4)$$

From equation (4) it can be observed that the FOC control of the self-excited generator can be implemented, via the linear dependency between machine torque and the i_{sq} current. The active current i_{sq} control loop in the d-q coordinates ensures a constant value for DC intermediate circuit voltage U_{dc} , while the reactive current i_{sd} is set to zero due to the rotor current creation of magnetic flux. In the investigated system, to obtain the rotational speed and stator field angle value which is needed for the Clarke and Park transformations, a sensorless system was used however, in many commercial applications, the use of an encoder is demanded in line with applicable rules and regulations.

The core of the FOC algorithm is the use of plane transformations calculated in real time between the three three-phase system and d-q plane. Utilizing space vector properties enables a sinusoidal projection of balanced three phase quantities as easy to control constant values of currents, voltages, and fluxes. For example, space vector x_s representing the quantities can be expressed by two-phase magnitudes, called x_α and x_β , in the real-imaginary complex plane.

Mathematically, this relationship can be written as:

$$\bar{x}_s = x_\alpha + j x_\beta = \frac{2}{3} (x_a + a x_b + a^2 x_c) \quad (5)$$

The α - β components of the resulting space vector can be calculated from the three three-phase magnitudes according to:

$$x_\alpha = \text{Re} \{ \bar{x}_s \} = \frac{2}{3} \left(x_a - \frac{1}{2} x_b - \frac{1}{2} x_c \right) \quad (6)$$

and

$$x_\beta = \text{Im} \{ \bar{x}_s \} = \frac{2}{3} \left(\frac{\sqrt{3}}{2} x_a - \frac{\sqrt{3}}{2} x_b \right) \quad (7)$$

Another very useful set of equations transforming phase quantities from the stationary abc reference frame to the d-q reference plane which rotates with the rotor is called a Park transformation can be expressed as follows:

$$\begin{bmatrix} x_d \\ x_q \\ x_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos\theta & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) \\ -\sin\theta & -\sin\left(\theta - \frac{2\pi}{3}\right) & -\sin\left(\theta + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \quad (8)$$

Equations given in (5–8) are hard coded into control software and executed in real time just to obtain values of currents

and voltages needed for easy control of the machine inverter and DC link voltage. For correct operation of the algorithm, a rotating frame angle θ calculation is necessary. In the steady steady-state working generator system presented here, the θ angle is calculated with the use of trigonometric functions and α - β plane voltages and currents derived from measurements and control system (voltages). Estimation of the rotor position angle can be estimated by means of integration of the speed or the use of machine state simulators and observers as well [15]. Due to the generating mode of operation, the parameters of generated voltage (frequency and voltages) must be known before the machine inverter can be put into operation and excite the alternator. It is important to match the initial frequency and amplitude of inverter voltage to the terminal voltage of the generator, thus preventing the flow of equalizing currents in the initial phase of synchronization. The simplest and most effective method of finding the frequency and rotational speed of a self-excited generator is to calculate of zero crossing events within a in certain time. This method is widely used in real world applications, but in the case of a synchronous generator co-working with an inverter it can be used only for a preliminary determination of voltage frequency. Because of distortions involved into the voltage values by an excitation rectifier working at low RPM values multiple zero value crossings may occur. In this case the analog or software AC voltage filtering should be applied, especially when the inverter excites the machine and controls DC link voltage.

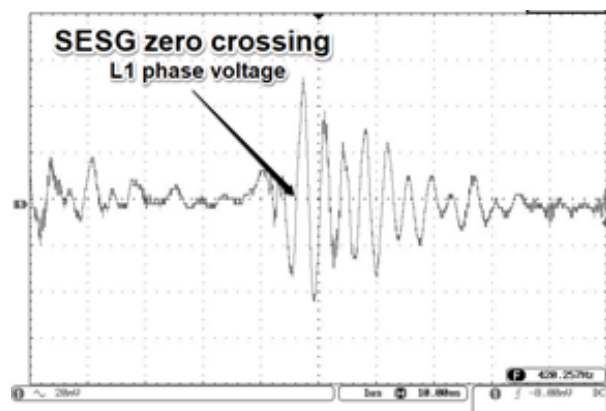


FIGURE 4. Magnified waveforms of phase voltage with multiple zero value crossings for a voltage frequency of 10 Hz.

A 7 kVA self-excited synchronous generator driven by a squirrel cage motor was used for purposes of this investigation. To obtain a suitable intermediate direct current voltage level, the electric generator must be controlled in a certain way. Because of self excitation and the use of a compound transformer to feed the voltage regulator in the synchronous generator, there is no need for independent control of the reactive current, therefore field oriented control (FOC) algorithm was chosen.

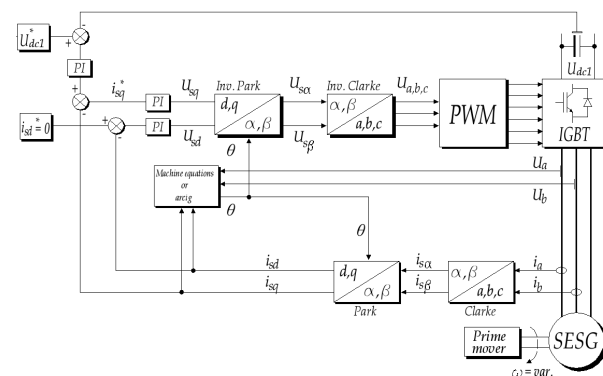


FIGURE 5. Scheme of the self excited synchronous generator FOC control structure.

The compound transformer is a very important part of the investigated system, as it assures a suitable excitation current, that is composed of the vector sum of the resulting phase voltages and currents.

Figure 6 presents an example of a brushless self-excited synchronous generator with a compound excitation system and a voltage regulator. The excitation winding of the synchronous generator receives power from the rotating rectifier, which is powered by a secondary coil of the compound current transformer.

The compound transformer contains a current coil (coil_I) and a voltage coil (coil_U). The sum of the both coils signals is created by the compound elements, which are in this case the capacitors. One compound element converts coil_U from a voltage source into a current source and shifts the phase (by about 90°) of the voltage signal. To ensure a sufficient initial self excitation of the synchronous generator, an additional coil (coil₀) powers the excitation coil (field winding) through a rectifier.

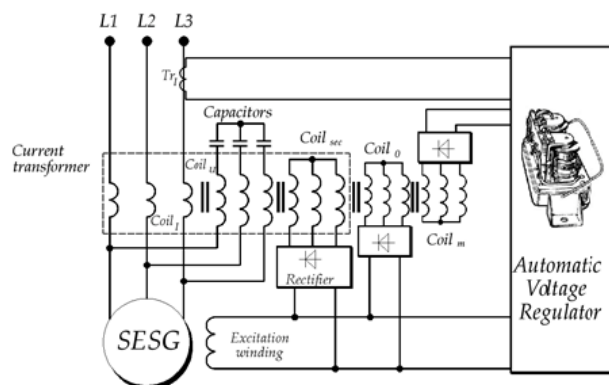


FIGURE 6. Scheme of an automatic voltage regulator with a compound transformer.

The compound transformer contains a current coil (coil_I) and a voltage coil (coil_U). The sum of the both coils signals is created by the compound elements, which are in this case the capacitors. One compound element converts coil_U from a

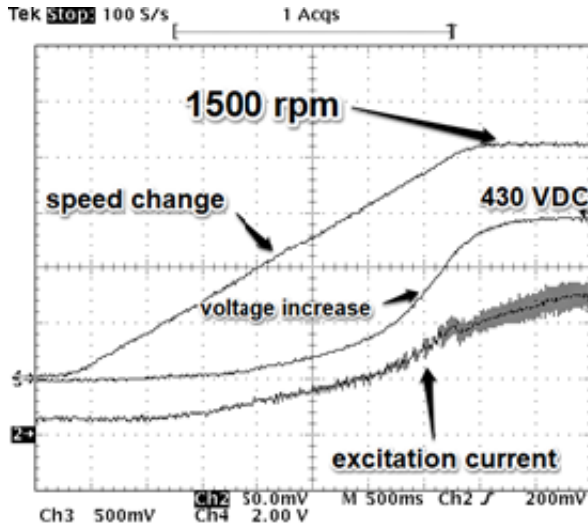


FIGURE 7. Nonlinear influence of rotational speed on field current, DC voltage and waveforms of the rectified SESG field winding current with no inverter action.

voltage source into a current source and shifts the phase (by about 90°) of the voltage signal.

The compound transformer is designed to provide supply a generator voltage ranging from an idle to an initial load. The automatic voltage regulator selects the redundant power above the nominal voltage, so in that most cases the voltage control is accomplished by reducing the excitation current. The simpler type of excitation unit was used in the proposed system. It does not contain neither additional winding (coil₀) nor an induction exciter with a rotating rectifier.

All excitation direct current flows through two slip rings and brushes from the voltage regulator into excitation windings. Nevertheless, it is the compound exciter which utilizes information of concerning currents, voltages, and phase shift to create a suitable, depending on load power factor rotor current. Because of the simple arrangement of the regulator presented here it is not recommended for parallel operation. Most of synchronous generators with regulators are designed for continuous work at almost constant rotational speed, thus a change of speed results in notable, nonlinear voltage changes. However, because of the DC voltage regulator, at some point (in this case from 260 V on the intermediate circuit) the process of excitation takes place and an increase to 560 V can be achieved just via i_{sq} current regulation. This property presents some disadvantage in a case where rotational speed was higher than nominal, because back emf generated by an alternator rotating at a higher angular speed will also be greater than nominal as the inverter pulses formed from higher intermediate circuit DC voltage.

B. ASYNCHRONOUS GENERATOR AND INVERTER SENSORLESS CONTROL

In the case of the asynchronous generator, which is a regular 5.5 kW squirrel cage motor the task of excitation and stable

work is far more complicated than in that of the SESG. For proper system startup, the DC intermediate circuit capacitors must be initially charged by an external source. To control an asynchronous generator in this manner, the algorithm prepared for the machine side inverter, supporting independent control of the magnetizing and capacitor charging current is necessary. The control method used in this case was a sensorless, multiscalar algorithm. The multiscalar control technique [16] utilizes variables like such as currents, voltages, and fluxes presented in the α - β plane, so there is no need of a Park transformation for control purposes. In contrast to a synchronous generator, almost every control scheme applied to an asynchronous generator uses a complex numerical machine model to operate it, because important values such as machine slip must be estimated. The accuracy of estimation of parameters depends mainly on machine parameters such as rotor time constant and especially rotor resistance which fluctuate according to machine load. Because of heating of rotor bars its resistance can change up to 20% and such a property can be considered as a drawback of FOC control method.

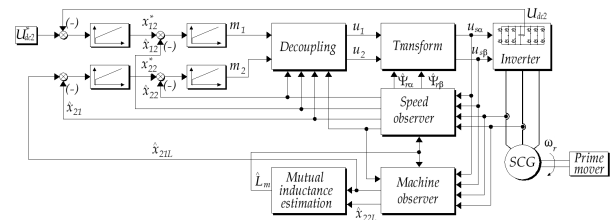


FIGURE 8. Nonlinear multiscalar control scheme of an asynchronous squirrel cage generator.

The set of asynchronous machine equations needed for vector control in the α - β plane is as follows:

$$\frac{di_{s\alpha}}{dt} = \frac{L_m}{\sigma L_r L_s} \omega_r \Psi_{r\beta} + \frac{R_r}{L_r} \cdot \frac{L_m}{\sigma L_r L_s} \Psi_{r\alpha} - \left(\frac{R_s}{\sigma L_s} + \frac{R_r(1-\sigma)}{L_r \sigma} \right) i_{s\alpha} + \frac{1}{\sigma L_s} u_{s\alpha} \quad (9)$$

$$\frac{di_{s\beta}}{dt} = \frac{L_m}{\sigma L_r L_s} \omega_r \Psi_{r\alpha} + \frac{R_r}{L_r} \cdot \frac{L_m}{\sigma L_r L_s} \Psi_{r\beta} - \left(\frac{R_s}{\sigma L_s} + \frac{R_r(1-\sigma)}{L_r \sigma} \right) i_{s\beta} + \frac{1}{\sigma L_s} u_{s\beta} \quad (10)$$

$$\frac{d\Psi_{s\alpha}}{dt} = -\omega_r \Psi_{r\beta} - \frac{R_r}{L_r} \Psi_{r\alpha} + \frac{R_r}{L_r} L_m i_{s\alpha} \quad (11)$$

$$\frac{d\Psi_{s\beta}}{dt} = \omega_r \Psi_{r\alpha} - \frac{R_r}{L_r} \Psi_{r\beta} + \frac{R_r}{L_r} L_m i_{s\beta} \quad (12)$$

$$\frac{d\omega_r}{dt} = \frac{L_m}{J L_r} (\Psi_{r\alpha} i_{s\beta} - \Psi_{r\beta} i_{s\alpha}) - \frac{T_L}{J} \quad (13)$$

where R_r , R_s are the generator rotor and stator resistances; L_s , L_r , L_m are stator, rotor, and mutual inductances, respectively; $u_{s\alpha}$, $u_{s\beta}$, $i_{s\alpha}$, $i_{s\beta}$, $\Psi_{r\alpha}$, $\Psi_{r\beta}$ are components of stator voltage, currents, and rotor flux vectors, respectively; σ is the leakage coefficient; ω_r is the angular rotor velocity; J is the moment

of inertia and T_L is the load torque. As it can be seen in (9) and (10), there are cross references between currents in the α - β planes. They can be eliminated via nonlinear control of the asynchronous generator. The coordinate change proposed in [16] for the rotor flux vector model transforms the induction motor model into a so-called multiscale model. For sensorless control system purposes, the transformation of the estimated stator current vector and air-gap flux vector to multiscale variables can be applied and it can be presented as follows:

$$x_{11} = \omega_r \quad (14)$$

$$x_{12} = \Psi_{ra}i_{s\beta} - \Psi_{r\beta}i_{s\alpha} \quad (15)$$

$$x_{21} = \Psi_r^2 \quad (16)$$

$$x_{22} = \Psi_{ra}i_{s\alpha} + \Psi_{r\beta}i_{s\beta} \quad (17)$$

where x_{11} is the rotor speed, x_{12} is the variable proportional to electromagnetic torque, x_{21} is the square of rotor flux, and x_{22} represents the so so-called magnetized variable [17]. The advantage of this method is that the generator torque is independent of the rotor flux position and stator current in any reference plane. The generator torque depends solely on the mutual reciprocal position of the vector flux and stator currents. In the case of a generator operating within a range from low to nominal speeds, errors arising from numerical integration of stator currents are small and, in practical solutions, can be disregarded, but in the case of higher speeds in a field weakening region, the system may get become unstable. For this reason, the presented system was supplemented by a disturbances observer which substituted numerical integrators thus reduced steady state errors in high speed region operation. The main goal of these calculations is determination of the rotational speed of the generator without the speed sensor which is fragile and susceptible to disturbances. The speed of rotation is needed for further subsequent rotor angle calculation and intermediate circuit DC voltage control. The value of the generator's rotational speed can be calculated by using the following formula, which includes the correcting coefficients ε_1 and ε_2 previously estimated in the speed observer:

$$\hat{\omega}_r = \frac{\varepsilon_1 \hat{\Psi}_{r\alpha} + \varepsilon_2 \hat{\Psi}_{r\beta}}{\hat{\Psi}_{r\alpha}^2 + \hat{\Psi}_{r\beta}^2} \quad (18)$$

Following an estimate of rotor speed and variables that control machine torque and magnetizing currents, regulator outputs are feed the decoupling equations for the system with variables designated m_1 and m_2 . The outputs of the decoupling system generate signals for the modulator controlling the SVPWM inverter. The error signal deriving from actual and set values of DC voltage comparison creates the coefficients needed in the x_{12} PI regulator just to maintain the required DC voltage value. The x_{22} variable depends on angular speed and changes in the field weakening region to obtain constant power.

Another important feature of SCG control software is the potential for flying start management. A flying start is available in the generating mode of an asynchronous motor when

the angular velocity is non-zero while the inverter is turned off. When it comes to switching the machine converter on, its initial frequency should match the rotational speed of the shaft. It is crucial to protect the system against the flow of equalizing currents of uncontrollable high values entailing protection action. Among drive techniques, there are known several methods of performing flying start, but the most appropriate in generating mode is use of a small amplitude alternating voltages which change over a wide frequencies range and to test the magnitude of the resulting current and intermediate circuit DC voltage.

The algorithm of generator's speed detection was set in a way such the initial DC voltage was applied at the beginning of the inverter speed search process and due to energizing windings, this voltage was decreased. As the frequency of inverter voltage approached the actual speed of the rotor, the intermediate circuit voltage uprose by several volts, and then dropped with a subsequent frequency increase. At the same time, when the machine slip was equal to zero, the phase current dropped to zero.

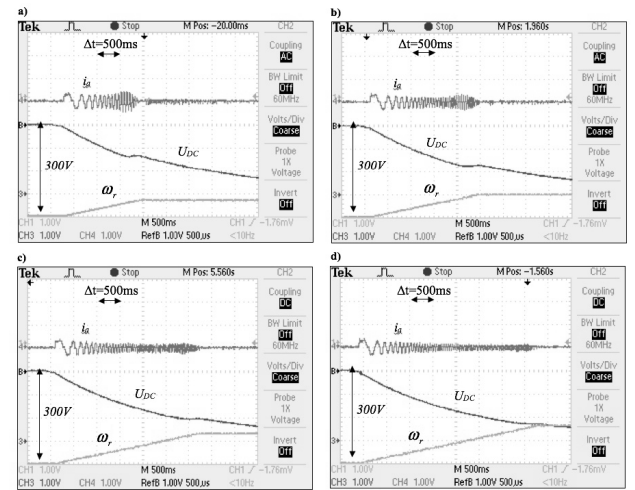


FIGURE 9. Waveforms of phase current i_a , intermediate U_{dc} voltage drop, and detected rotational speed ω_r for speeds of: 126 rad/s (a), 189 rad/s (b), 251 rad/s (c) and 314 (d) rad/s (d).

A change in the sign of DC voltage rate expansion along with zero phase current value informs the user of actual rotational speed and is forwarded to inverter controls to initiate the process of power build-up. Because some fluctuations in the speed may occur during the detecting process, the subsequent excitation process implements initial current and voltage limitations and thus, even in the presence of disturbances, inverter currents do not exceed safe values.

C. CONTROL OF COMMON INTERMEDIATE CIRCUITS DC VOLTAGES

The auctioneering diodes (known also as DOring or ORing diodes) are widely used to isolate redundant power sources so that a single source failure will not bring down the entire system [7]. In the presented system diodes are necessary to

avoid flow of circulating currents between machine windings and inverters but despite the popularity of this solution, there are several problems associated with this kind of isolation system. One of which is the loss due to the insertion of a semiconductor device with some degree of forward voltage drop directly into the main power path. The full load current flows through diodes and the associated power loss, dissipated in the form of the heat which is load direct current multiplied by the direct voltage drop across the device. In the case of a power grid system, the amount of power is notably high, and along with reduced the efficiency of the whole system, heat is generated in the diode structure.

The other issue associated with the use of auctioneering diodes is the reverse recovery time. When a short circuit occurs in one of the sources, the device must limit the flow of high current within a short period of time. If not, the bus can be degraded to the point where the load will fail due to undervoltage or transient conditions.

Similar systems incorporating auctioneering diodes are widely used to distribute power between renewables and their power converters. The combination of parallel connected voltage sources acts as one large supply with controlled load on each of the units. The reliability of the whole system can be secured by taking advantage of load sharing to incorporate modular redundancy.

One of the main problems in the control of such devices is the distance between sources and the resistance of the power lines connecting converters. Lines of unequal lengths cause uneven resistances, what has a significant influence on measurements, and can may be cause of unstable operation of the system, in connection with the droop voltage control method. In the presented system, the generator-inverter sets are close to each other and power lines are of the same length thus the problem of uneven DC voltages measurements is negligible.

As it was mentioned above to prevent the flow of circulating currents through conducting transistors of the SESG inverter and SCG converter connected in parallel, the auctioneering diodes must be implemented into in the power circuit. Regarding power dissipation and thermal losses, there are some alternatives to plain, high voltage diodes, such as Schottky diodes or faster power MOSFET units. In this system, both of generators and their inverters are controlled independently, the switching patterns are different, so turn-on states of the transistors can occur simultaneously.

In this case, the equalizing currents will flow and control of power distribution between generators will be inaccessible.

Placing only one diode on the positive output of each intermediate circuit leg would be sufficient to prevent the flow of circulating and equalizing currents. Unless a negative diode is inserted, a circulating current occurs and problems in power sharing are unavoidable.

In the presented system, auctioneering diodes are placed in series with both the positive and negative output feeds. The greatest disadvantage of this solution is a ground fault scenario involving both positive and negative inverter legs,

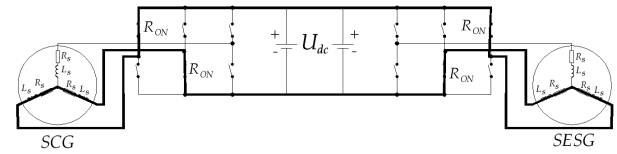


FIGURE 10. Flow of circulating currents in the absence of auctioneering diodes in the main circuit path.

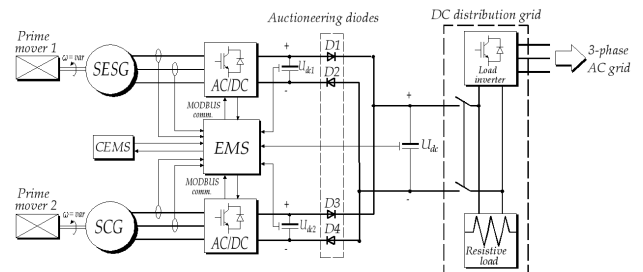


FIGURE 11. DC grid connected through auctioneering diodes to the intermediate circuits of parallel- operating machine-inverter M11 and M12 sets. CEMS – combustion engines management system.

leading to double voltage stresses on loads during a dual ground fault, which in turn must be managed by the inverter control system and protections [2].

There are several schemes designed to achieve load sharing between electrical voltage sources. The most common passive and active techniques presented in the [18].

One of the techniques is the droop method, which controls the converter output impedance to achieve load sharing at a suitable ratio. Because it is a simple open loop method, it is not very accurate.

The use of a series connected ORing diode, which is the passive method and constitutes the most common control of paralleling DC voltage sources. By using converters with adjustable outputs, it is possible to share a certain amount of current between power electronic devices. This method has the inherent disadvantage of leading to an additional power loss in the diodes as the generated heat.

A popular master – slave configuration, whereby one of the units is selected as a “master” module and its main task is the voltage control, but also forces the remaining modules (slaves) to act as current sources by communicating and sending control commands. This approach, with current mode supplies, will facilitate current sharing but it will not achieve redundancy. In such case an additional external controller performs the task of load sharing. All sharing signals (e.g. currents) are compared to the individual converters in the controller, which adjusts the corresponding feedback signal to balance the load currents. In its simplest form, this system requires an additional control module with suitable inputs and electrical connections between the controller and each of the parallel connected sources.

The automatic current sharing system requires an adjustment amplifier that compares a current signal flowing from

the common DC bus to the current of the individual units and adjusts the reference of the voltage signal until equal load current distribution is achieved.

The power power-sharing techniques are suitable for the units supplying the systems in a continuous and reliable manner. The methods of implementation of power sharing realization are comprehensively described in [19] but given the autonomous local control of the voltage sources, the focus was on droop methods. The power distribution is an important part, required to improve the operation (reduce thermal stresses, etc.) and to secure an uninterrupted power supply. In the presented system, power sharing utilizes some of these elements but, by virtue of its functions, differs in concept and design. The main purpose of the investigated system is to distribute power among the generators and inverters as specified by the electrical management system based on the signals analysis obtained from the prime mover management system. This task requires an external module that communicates with external units and sends control signals to each converter. These control signals contain an optimal power load for rotating generator set that varies in time. Because a reliable power plant must be equipped with at least two independent electrical power sources, then the external controller collects data from every unit, combines them into a complex algorithm, and controls power sharing over all generator sets. The communication time lags between the controller and converters control units in the most of practical cases is not crucial, since every inverter has its own real time control system, designed to be fail safe. In the case of communication failure or the occurrence of heavily disturbed signals the system maintains the state of actual outputs until functionality is restored. The classic electrical plant system used on a board of the ships operates inherently in the master-slave mode during parallel operation. Power is distributed proportionally to the instantaneous power of the synchronous generator, which is affected mainly by the working conditions of the combustion engine. The main alternator set, as “master” maintains the frequency and voltage amplitude, while the slave generator works in active and reactive power deliverance mode. In the stand-alone operation mode, one generator acts as a “master”, supplies the busbar with voltage of power grid frequency and amplitude. The discussed system works in a similar way, but the only kind of controlled power sharing derives from direct voltage and current real-time analysis. Although the control system works on signals obtained from the DC side of the inverters, it affects on variables controlled by machine side inverters connected to three phase generators terminal.

The proposed system contains an external unit called the energy management system (EMS), consisting of DSP and FPGA circuitry, which in real time acquires measurements data from both machines inverters and their intermediate circuits. Apart from electrical quantities measurements, the EMS exchanges data with the combustion engine management system (CEMS). Incoming data from the CEMS are needed to perform control tasks, mainly on the generators rotational speed, to achieve optimal fuel consumption by

the diesel engines. The communication between the CEMS, EMS, and inverters is performed using a Modbus protocol where the EMS is set as the Modbus master and the inverters control units of the inverters are Modbus slaves. The control algorithm and dependencies between the CEMS and EMS, along with the logic behind them, are beyond the scope of this article.

To establish suitable active power sharing between generator inverter sets, some means of controlling output current and DC voltage must be provided. Such a system would work by controlling intermediate circuit DC voltage and limiting the resulting power by restricting the converter output current according to energy management control signals. In the simplest form, the power sharing algorithm controls the voltages of common bus U_{dc} , SESG bus U_{dc1} , and SCG bus U_{dc2} , along with the corresponding inverters currents.

When one of the generators is set as the master, its voltage U_{dc1} is maintained for the DC common bus while the current limitation is set to the nominal value of the inverter-generator set.

When the second generator is put into operation, on the idle run, its DC voltage U_{dc2} is kept below bus voltage U_{dc} , and the generated current is only a small active current that covers the electrical and mechanical losses. When the power sharing signal is sent from the energy management system, the DC voltage regulator of the slave inverter increases U_{dc2} thus commanded voltage extends the bus voltage U_{dc} , the unit takes over some part of the load current. The amount of this current depends on the forced voltage thus it can be stated that with higher generated voltage, the greater load is taken by the inverter. Because of the impulse-like nature of currents charging intermediate circuit capacitors, which causes floating DC voltage value, the auctioneering diodes are switching on and off all the time continually when in power sharing mode, which is the main reason for increased heat dissipation. The silicone-based diodes are characterized by a negative temperature coefficient therefore, without a current limiting a load increase would occur.

Differences in the voltage values during power distribution are slight, ranging from a several to a dozen volts, as can be observed in figure Fig. 13. Therefore, to achieve precise management of load sharing, the control system was altered to accordance with the layout presented in the Fig. 12. To improve the accuracy of current sharing, an additional PI regulator was added to each converter control loop. In the main algorithm, inverters are carry out the task of generating the DC voltage at a constant value, which is fixed in the control program. As for power sharing, a small amount of voltage is added or subtracted from the no-load value. The additional regulator takes the commanded current value and subtracts the actual active current. An error signal feeds the regulator, which in turn produces an extra voltage signal which is added to the main loop voltage regulator. Instead of this, an adaptive regulation can be applied to voltage controllers in the main program loop, although from a practical point of view it is more convenient to have an independent

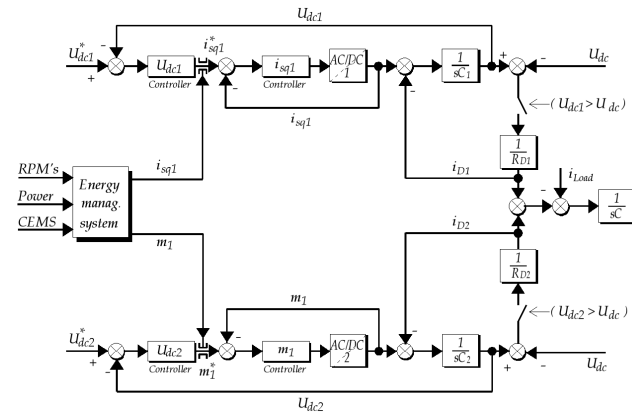


FIGURE 12. Control scheme of power sharing algorithm between two voltage sources with current limiting and auctioneering diodes.

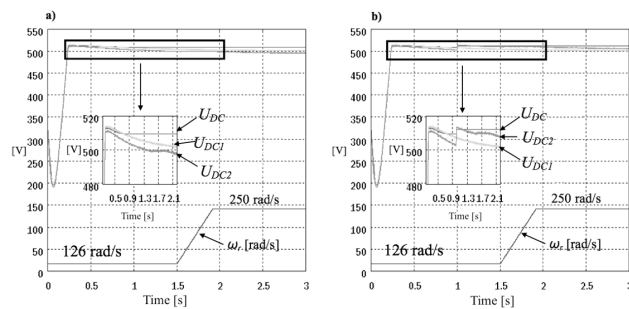


FIGURE 13. Simulation results of DC voltages in power sharing between two inverters fed by asynchronous generators [20].

regulator just exclusively for slight corrections of current and voltage values.

II. EXPERIMENTAL RESULTS OBTAINED AT A LABORATORY STAND

To verify the validity of the adopted assumptions and the created method of power sharing, the proposed system was subjected to laboratory tests. The simplified layout of the presented system is shown in the Fig. 11. Instead of combustion engines driving both generators, inverter supplied induction machines were utilized. This solution made it possible to observe and control the power flow and precisely record the input parameters, with a special focus on incoming torque and angular speed. Communication between the modules was carried out using the standard Modbus protocol, with optical isolation on each inverter to minimize the number of false frames caused by electromagnetic interferences. For the load of the intermediate circuit, high power resistors, along with a commercial inverter with 4 kW of maximal power, with an induction motor were used.

This power distribution system was tested by examining the quality and stability of the voltage generated by gensets with converters individually, both at idle and under load, at varying rotational speeds. Because of the specific features of the SESG and SCG used in the experiments, the initial

conditions differed between cases. The control of the self excited generator and associated inverter was based on initial charging of intermediate capacitors to the minimal voltage required for proper FOC algorithm operation. To control voltage level and bypass the capacitors charging unit, a contactor was used. The start of the SESG control algorithm is also of the flying start type accordingly system first detected the frequency of angular speed and then forced the inverter to supply the windings with aligned frequency voltages. The magnitude of commanded voltages has increased with the use of limiting ramps, resulting in a smooth “synchronization” process with no occurrences of uncontrollable current peaks or voltage spikes. When the initial process was complete, the inverter generated the voltages which produced the load angle with rotating machine back emf, and DC voltage reached the desired value of 560 V.

During the test, a second inverter DC bus was disconnected, and the resistive load was applied to the SESG inverter. The processes of load application and shedding were resulted in several rapidly decaying voltage oscillations. The value of the active current in the presented control technique depends on the angular speed of the generator due to a constant amount of power. The inverter current value was increased with a drop in speed and reduced when speed increased.

The next tests were performed to check the asynchronous generator in an island operation. The initial switching on of the SCG inverter was preceded by charging the capacitors from rectified 3-phase AC voltage to 600 volts. Following completion of this process and initial rotating machine “excitation” performed by the inverter charging circuit was bypassed with the contactor.

Active power covering losses during an idle run is provided by an excited machine and created by inverter slip, whereas the capacitors are the source of the reactive power magnetizing the asynchronous generator. To ensure proper operation of the generator, the DC voltage was maintained at a constant level of 560 volts. As it can be seen in Fig. 15, under load operation the control system of SCG keeps the DC voltage tight, but some correction of coefficients is needed due to current ringing during no-load operation. Just for the sake of the simplicity, the currents in the α - β reference frame were

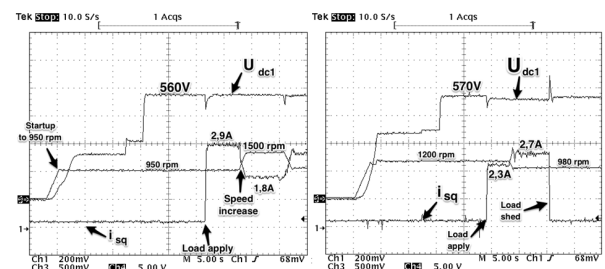


FIGURE 14. Waveforms of DC voltage, active current, and rotational speed with 1.5 kW load apply application and load shedding of a stand-alone, self excited synchronous generator while during angular speed changes.

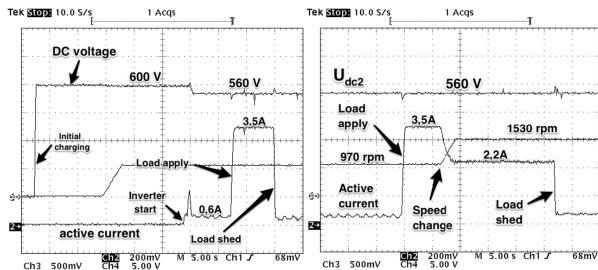


FIGURE 15. Waveforms of DC voltage, active current, and rotational speed with 1.5 kW load apply application and load shedding of the squirrel cage generator during rotational speed changes in island operation.

transformed into the d-q constant in the time values plane. This procedure has enabled unified presentation of the results and had no influence on system performance.

To prove the robustness of the proposed sharing current-sharing algorithm with an additional PI voltage controller, the aforementioned different types of generators and accompanying converters were connected through auctioneering diodes and DC current contactors to a common bus. The DC voltages on both machine inverters buses were kept at the same value of 560 V in the island mode operation. First, the resistive load was applied to the SCG which was set as a master in the test. Subsequently, the squirrel cage generator inverter intermediate circuit was connected to the common bus and placed into power sharing mode. The DC voltages has evened out, and the additional PI regulator embedded in the SCG control unit pulled up the voltage value to the active current limit set by the EMS.

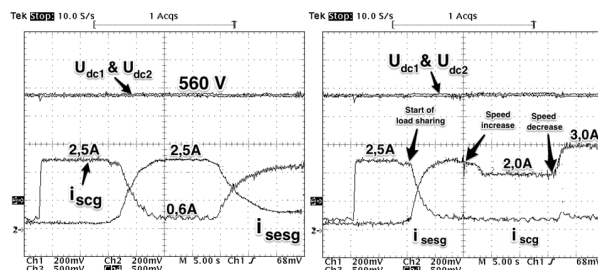


FIGURE 16. Waveforms of DC voltage, active current, and rotational speed with 1.5 kW load apply application and load shedding of a squirrel-cage generator during rotational speed changes in island operation.

As it can be observed in figure Fig. 16, the current limit was set so all the power was taken by the slave generator. After a set time, the slave alternator and its inverter dropped 80% of the load, which what caused increasing an increase in the load share of the master generator.

Since the main idea behind the presented active load sharing technique concerns the potential in generator speed change, the tests showing this feature were carried out. The angular speed of the slave generator with a load taken from the master was changed along with a simultaneous change in

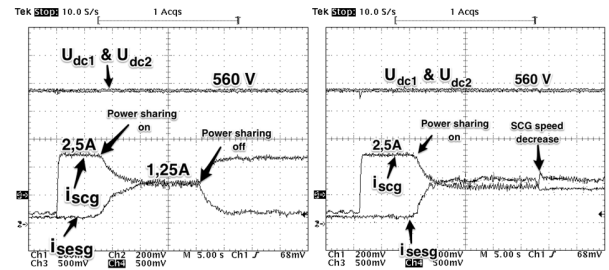


FIGURE 17. Waveforms of DC voltages and active currents with 1.5 kW load apply application and load sharing between the SESG and SCG in constant RPM operation and with speed change.

current, as it can be observed in figure Fig. 17, the current was changing simultaneously.

Subsequent tests proved that the proposed method enables partial power sharing between generators and inverters in a controlled way. The generators were put into parallel operation, in which the consumer current was set to load equally both of generators. In a long time, stable work operation of the system was achieved, and the control unit was able to maintain the assumed load share ratio even during load changes. The speed change of the slave generator during the parallel power sharing operation resulted in current change as it was in stand-alone generator operation, what can be seen in the Fig. 17. Because the rotational speed may range from 50 up to 100% of the nominal value, the system must be ready to limit requested power, especially during low speed operation. As can be deduced from the experiments results, in the low speed region both the machine and converter currents can reach the nominal value.

The proposed additional voltage regulator with limited current output constituted a good response to the power sharing demand and its regulating accuracy reached 95% in steady state operation. The accuracy of power sharing depends mainly on voltage control of the DC intermediate circuit and even small changes in this voltage will cause notable changes in power distribution. Operation over a long period of time has proved that the system behaved in a stable manner within the assumed range of changing speeds range and with a variety of electrical loads, including switchable resistances and inverter drive. The rate of load changeover can be tuned by means of the appropriate setting of PI regulator coefficients. In the presented system the settling time was set at 2.5 s.

III. DISCUSSION

As the experimental results show, a system consisting of a synchronous and a squirrel cage generator can operate in parallel for a long period of time, if it is equipped with power electronic converters. An appropriate initial excitation and robust control technique enables the production of electrical power with the use of a variable speed generator. The useful speed range of generators for correct operation varies from 40 to 100 %. When the synchronous generator was used

modulator were sent from the processor in the program interrupts. The programming of FPGA was performed with the use of Quartus II Web Edition software, whereas DSP software was written and compiled into a hexadecimal loader file in VisualDSP++ 5.0 IDE. The energy management board was of the SH363 type and consisted of two boards. The FPGA board has also possessed input/output port which was electrically compatible with an industrial Modbus RS485 two-wires port. Through optically isolated RS485 ports, EMS sent and received control commands to slave converters and received data from combustion engine control units. Due to its relatively slow transmission, the Modbus was not fit for real-time fast control tasks, but its capabilities were sufficient for the transmission of information in non-critical systems such as combustion engines management. In order to obtain simultaneous oscillograms, each unit was equipped with two 8 bits digital to analog optoisolated converters enabling the real time variables presentation.

All of the processor boards were controlled in real time by means of dedicated software communicating with PC computers with the standard USB 2.0 ports.

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Authors' photographs and biographies not available at the time of publication.

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